

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
Lecture: CPR 37

**Ventilation Perfusion Inequality and
Hypoxemia**

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Objectives

By the end of this lecture, students should be able to:

SOM.MK.II.BPM1.3.CPR.2.PHYS.0246	Define the ventilation/perfusion (V/Q) ratio of an alveolar-capillary lung unit and show how it determines the PO ₂ and PCO ₂ of the blood emerging from that lung unit.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0247	Calculate the average V/Q ratio in a normal lung and explain how V/Q is affected by the vertical distribution of ventilation and perfusion in the healthy lung.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0248	Predict how the presence of abnormally low and high V/Q ratios in a person's lungs will affect arterial PO ₂ and PCO ₂ .
SOM.MK.II.BPM1.3.CPR.2.PHYS.0249	Describe two causes of abnormal V/Q distribution.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0250	Explain the right-to-left shunts, anatomic and physiological shunts, and physiologic dead space (wasted ventilation) and the consequences of each for pulmonary gas exchange.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0251	List and explain the airway and vascular control mechanisms that help maintain a normal ventilation/perfusion ratio. Name two compensatory reflexes for V/Q inequality.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0252	Calculate the alveolar to arterial PO ₂ difference, (A-a) O ₂ gradient. List the normal value range for (A-a) O ₂ and explain the significance of an elevated (A-a) O ₂ .
SOM.MK.II.BPM1.3.CPR.2.PHYS.0253	Describe how changes to the A-a gradient occur and what factors lead to an increased A-a gradient in patients.
SOM.MK.II.BPM1.3.CPR.2.PHYS.0254	Describe the mechanisms for the shift in alveolar ventilation that occur immediately upon ascent to high altitude, after remaining at altitude for two weeks, and immediately upon return to sea level.

Assigned Reading

Rhoades and Bell, *Medical Physiology: Principles for Clinical Medicine*, Sixth Edition, © Lippincott, Williams & Wilkins (2023), Chapters 20, 21

- <https://meded-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=253088770&bookid=3211>
 - BLOOD FLOW DISTRIBUTION IN THE LUNGS
 - SHUNTS AND VENOUS ADMIXTURE
- <https://meded-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?bookId=3211§ionId=253088906&resultClick=1#253089023>
 - CONTROL OF BREATHING IN UNUSUAL ENVIRONMENTS

Important Formulas

1. V_A/Q ratio = Alveolar Ventilation/Cardiac Output
2. **Transpulmonary pressure (PTP) = (Alveolar Pressure) - (Intrapleural Pressure)**
3. **A-a O₂ Gradient = $P_AO_2 - P_aO_2$**

SOM.MK.II.BPM1.3.CPR.2.PHYS.0246

Define the ventilation/perfusion (V/Q) ratio of an alveolar-capillary lung unit and show **how it determines the PO_2 and PCO_2** of the blood emerging from that lung unit.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0247

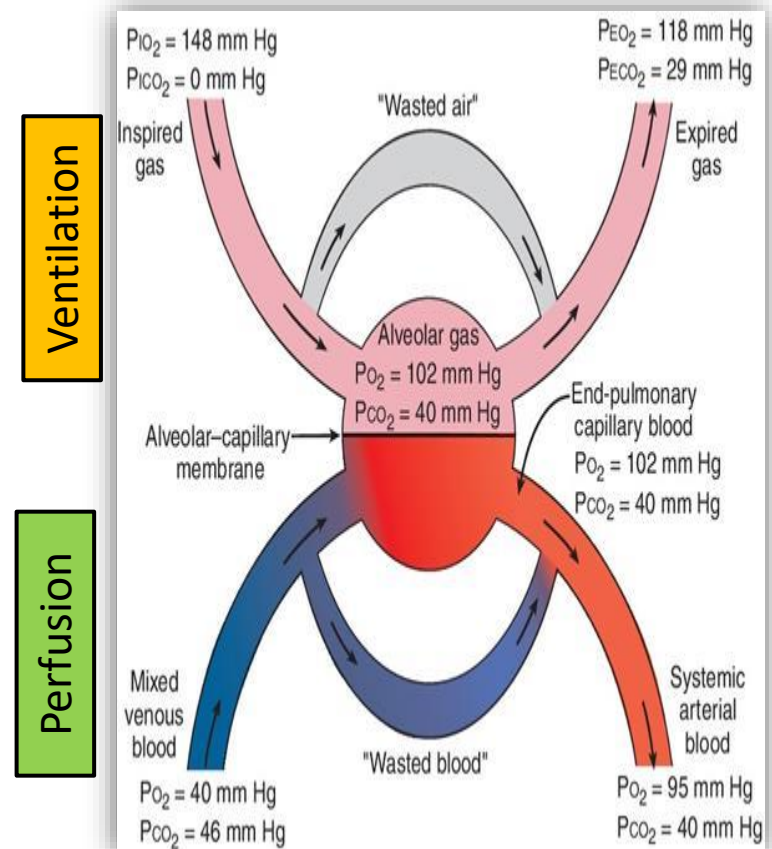
Calculate **the average V/Q ratio in a normal lung** and explain how V/Q is affected by the vertical distribution of ventilation and perfusion in the healthy lung.

Ventilation Perfusion Ratio

- Alveolar Ventilation (V_A) – 4L/min at rest
- Perfusion (Q) – cardiac output 5L/min
- Ideal V_A/Q ratio – $4/5 = 0.8$
- The V_A/Q ratio compares alveolar ventilation with blood flow in lung regions.

Ventilation Perfusion Ratio and Hypoxemia

- Alveolar-arterial oxygen (**A-a O₂**) gradient – 5 to 15 mm Hg is due to venous admixture.
 - Shunts
 - Variations in V_A/Q ratio
- Increased venous admixture results in low O₂ tension in the blood (hypoxemia)



SOM.MK.II.BPM1.3.CPR.2.PHYS.0249

Describe two causes of abnormal V/Q distribution.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0250

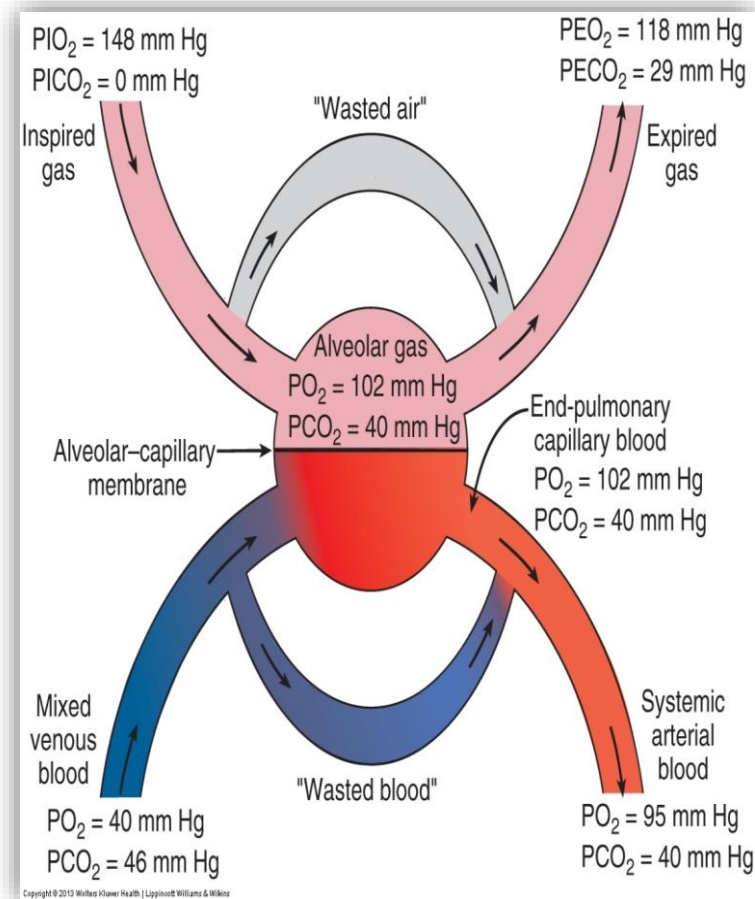
Explain the right-to-left shunts, anatomic and physiological shunts, and physiologic dead space (wasted ventilation) and the consequences of each for pulmonary gas exchange.

Shunts and Venous Admixture

- There is imperfect matching of lungs' airflow and blood flow
- **Physiologic Shunt** (total venous admixture) = venous admixture from **anatomic shunt** + venous admixture from low V_A/Q regions.
 - i) 50% comes from **anatomic shunt (right -to -left shunt)** (bronchial circulation, thebesian veins)
 - ii) 50% from **low V_A/Q** e.g. at the base of the lung, partially obstructed airway

“Wasted air” and “Wasted blood.”

- All of the inspired air does not participate in gas exchange, resulting in some “wasted air.” – **physiologic dead space.**
- All of the blood entering the lung is not fully oxygenated, leading to some “wasted blood.” - **physiologic shunt.**



Venous Admixture and Shunt

An anatomic shunt (*right-to-left shunt*) has a structural basis and occurs when blood bypasses alveoli through a channel. Examples include:

1. Right to left heart through an atrial or ventricular septal defect
2. From a branch of the pulmonary artery connecting directly to the pulmonary vein.
3. The bronchial circulation also constitutes shunted blood because bronchial venous blood (deoxygenated blood) drains directly into the pulmonary veins, which are carrying oxygenated blood.

“Wasted air” and “Wasted blood.”

TABLE 20.1

Shunts and Dead Spaces Compared

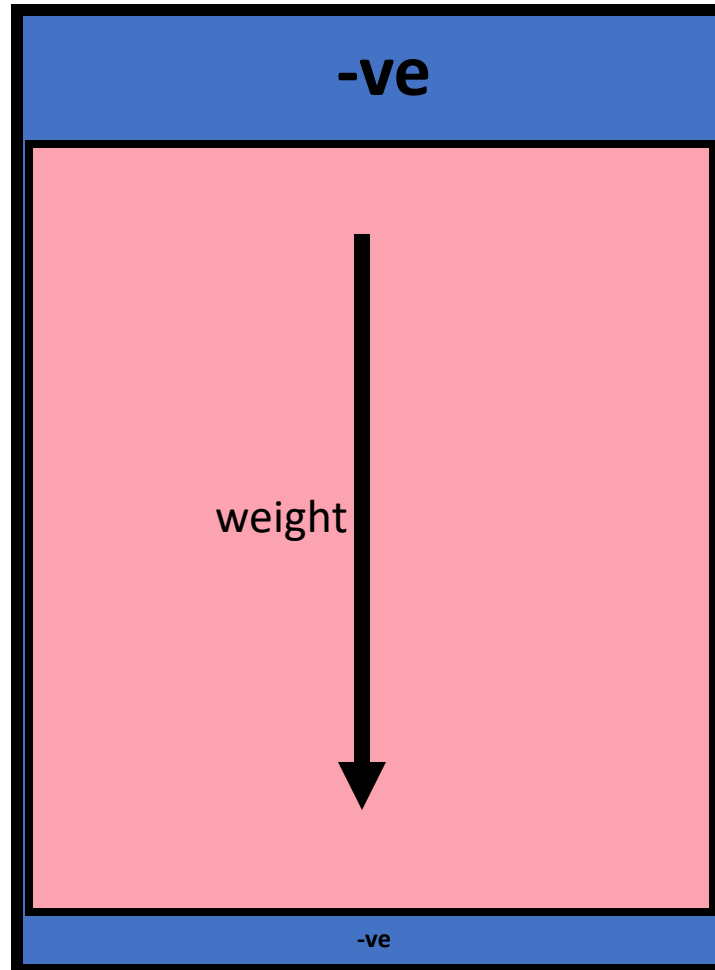
Shunt	Dead Space
Anatomic	Anatomic
+	+
Low ($\dot{V}_A/\dot{Q}$) ratio	Alveolar
=	=
Physiological shunt (calculated total “wasted blood”)	Physiologic dead space (calculated total “wasted air”)

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SOM.MK.II.BPM1.3.CPR.2.PHYS.0247

Calculate the average V/Q ratio in a normal lung and explain how V/Q is affected by the vertical distribution of ventilation and perfusion in the healthy lung.

Lungs are Heavy



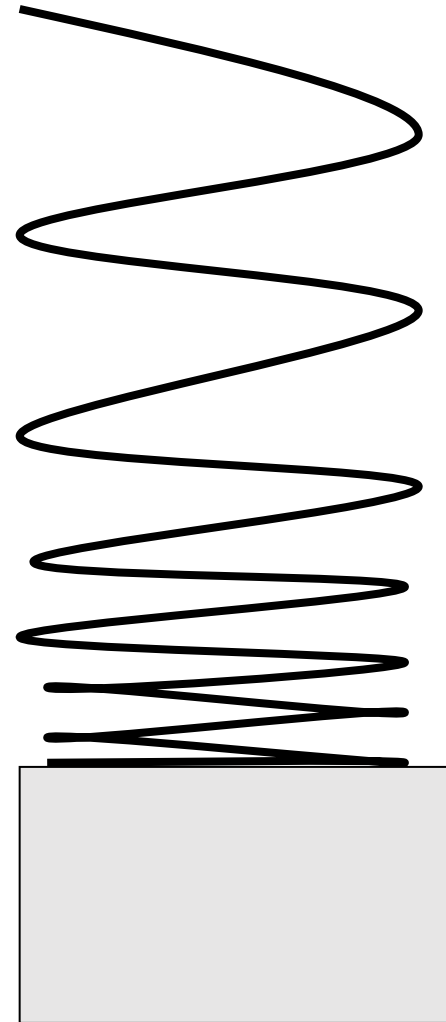
Lung is a Slinky

TOP – has most of the weight of the rest of the slinky pulling down on it – therefore the coils are pulled apart

MIDDLE – has half of the mass of the rest of the slinky pulling down on it – therefore the coils are tighter together than at the top

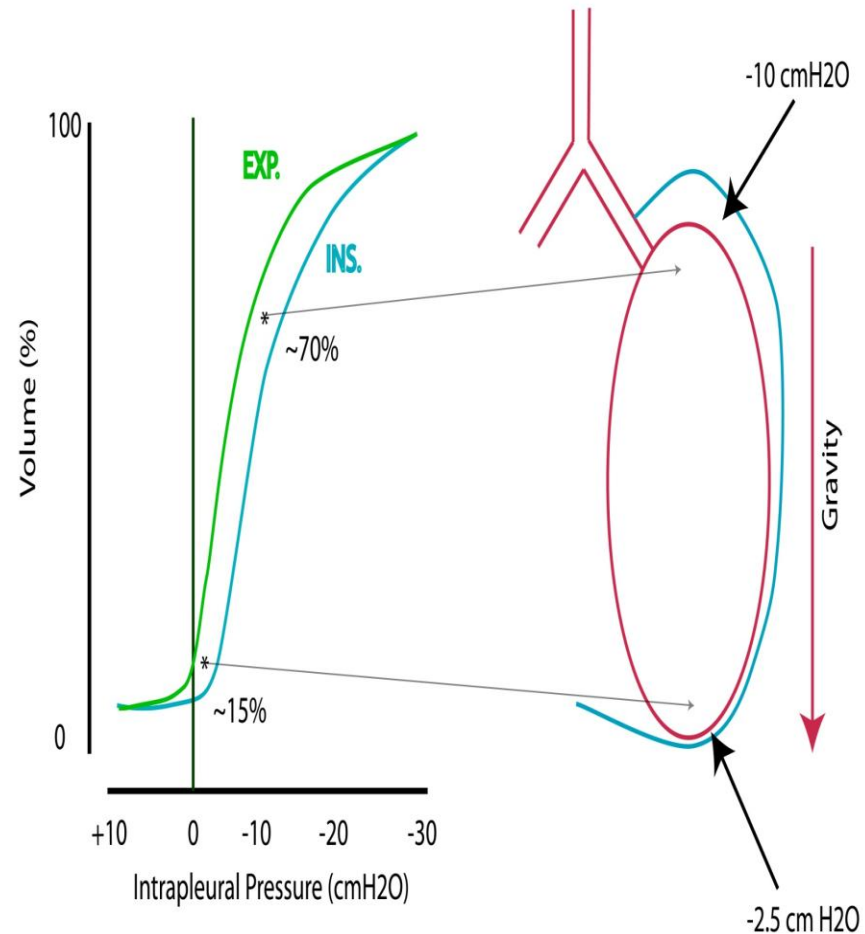
BOTTOM – has very little weight of the slinky pulling down on it – therefore the coils are tighter together than at the top

In addition the lung is resting on the contents of the abdomen



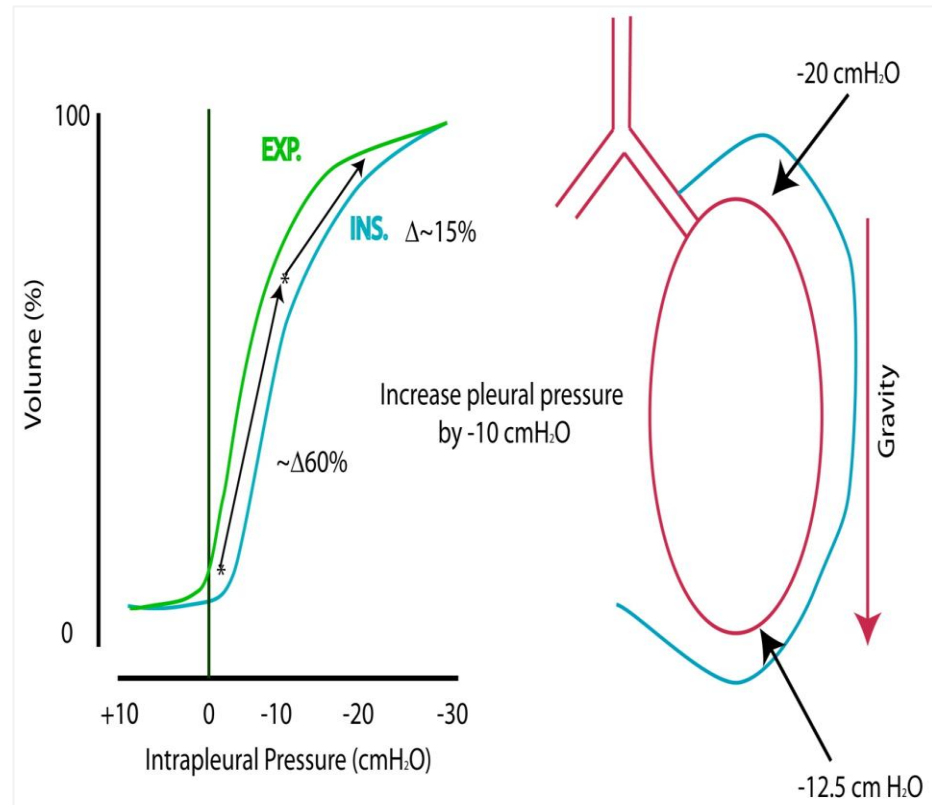
Pressure Volume at FRC

- The apex of the lung is less compliant than the base.
- At rest the lung apex is approximately 70% distended.
- At rest the lung base is 15% distended.

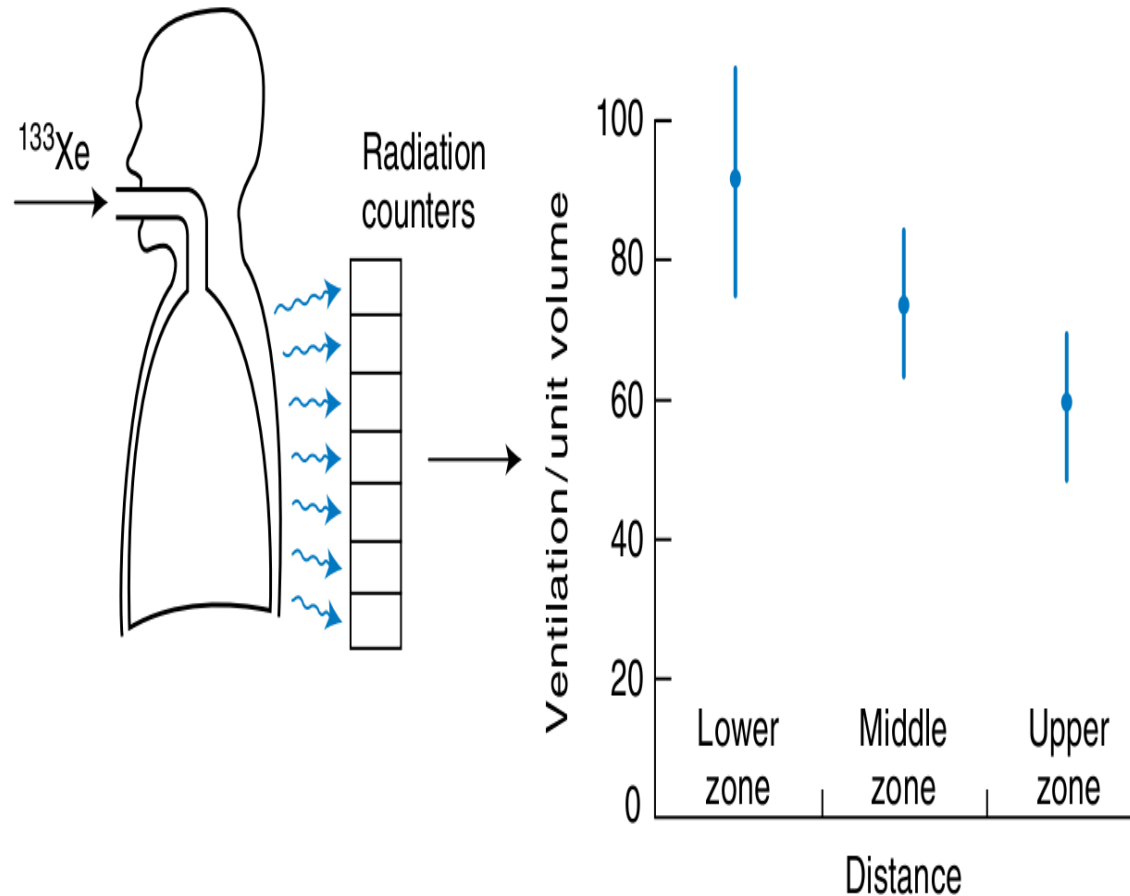


Pressure Volume Changes During Inspiration

- During inspiration the change in volume at the apex is significantly less than at the base.
- The ventilation gradient is aligned with gravity.

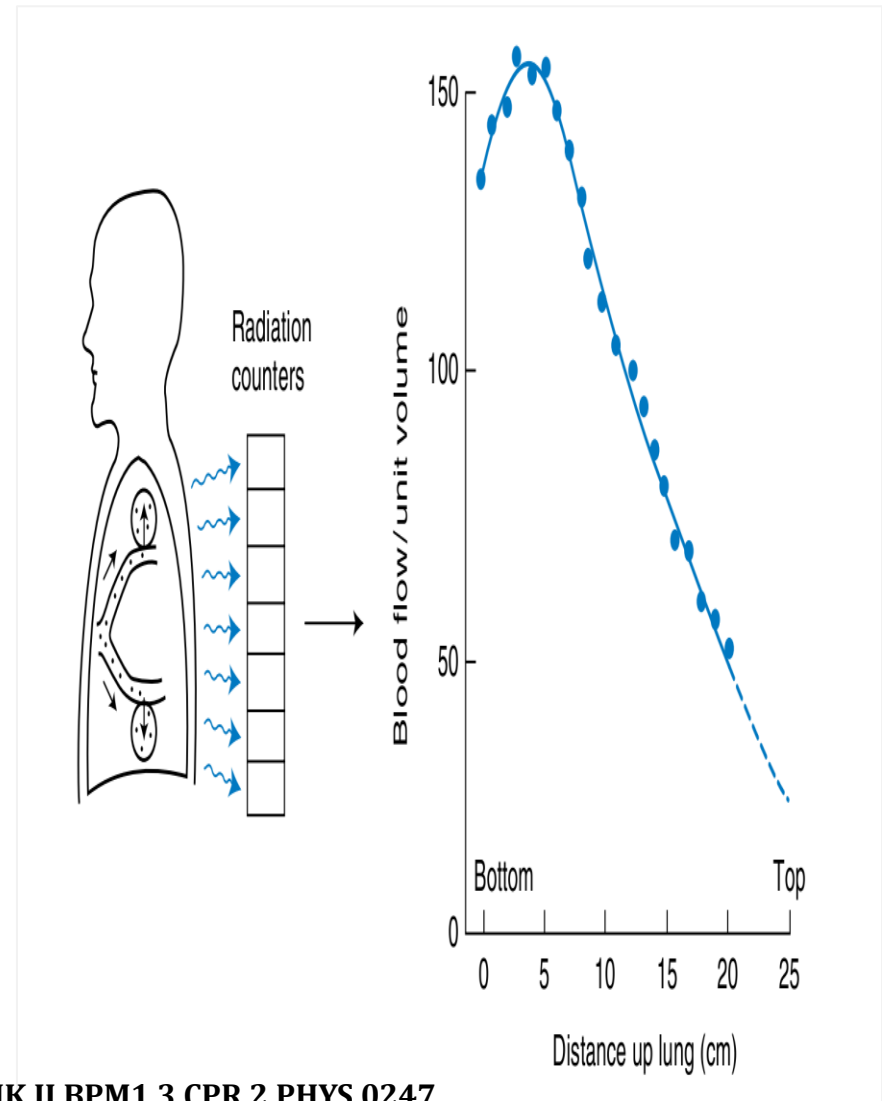


Regional Differences in Ventilation



Regional Differences in Perfusion

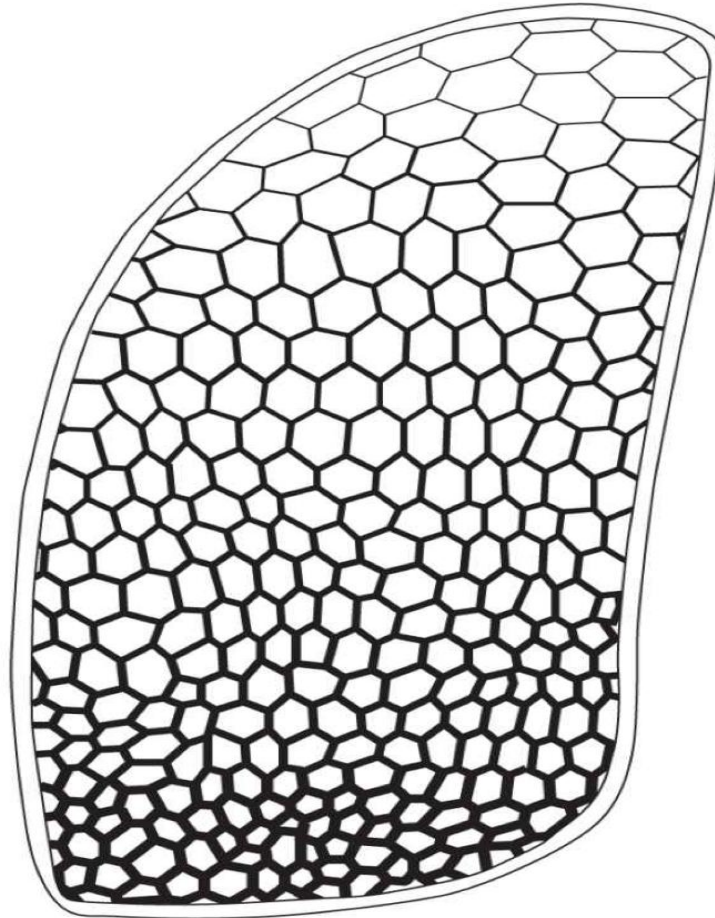
- Pulmonary circulation is a high-flow, low-pressure, low-resistance system.
- Since upward flow runs against hydrostatic pressure there is more resistance to blood flow toward the apex of the lung (no hydrostatic pressure to overcome in blood flowing down)
- Alveolar pressure also affects pulmonary perfusion – high alveolar volume in the apex reduces blood flow (this restriction is not present when alveolar pressure falls below pulmonary arterial pressure)



Ventilation & Perfusion: Regional Differences

Ventilation
Intrapleural pressure
more negative
Greater transmur-
al pressure gradient
Alveoli larger, less
compliant
Less ventilation

Intrapleural pressure
less negative
Smaller transmur-
al pressure gradient
Alveoli smaller, more
compliant
More ventilation

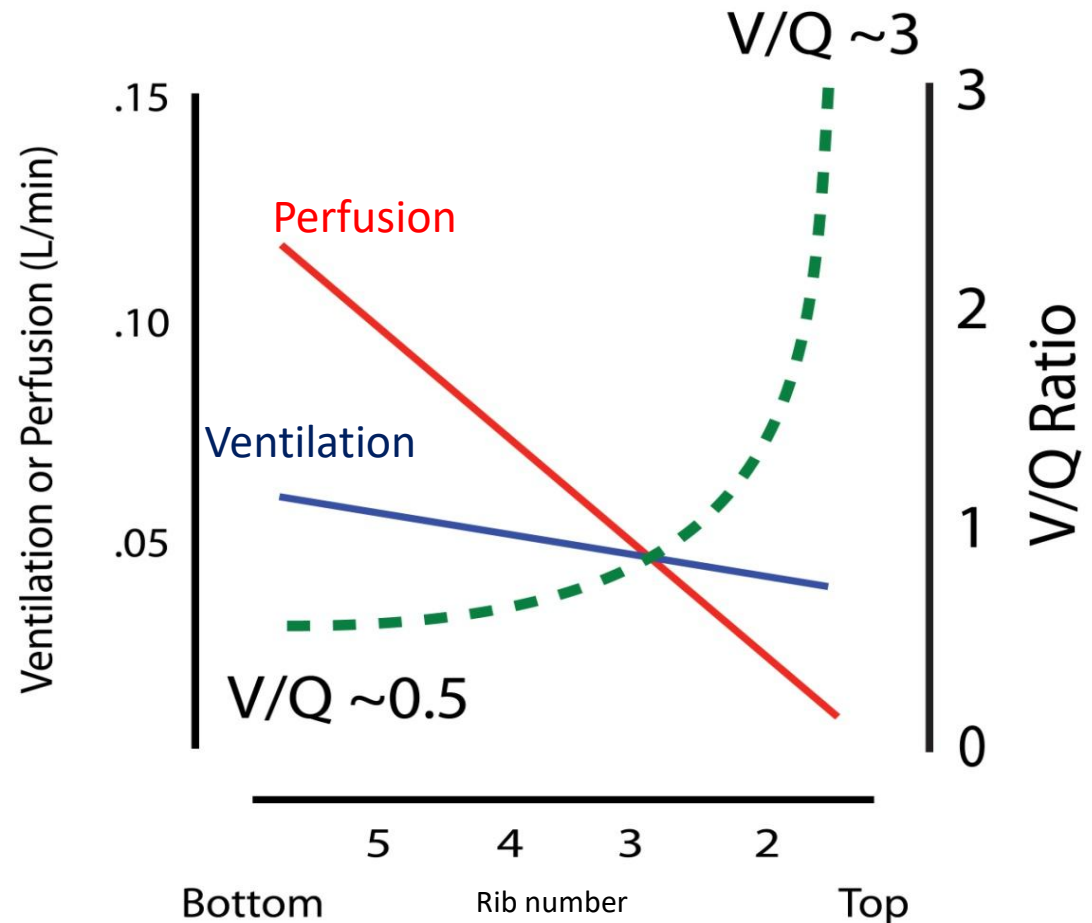


Perfusion
Lower intravascular
pressures
Less recruitment,
distention
Higher resistance
Less blood flow

Greater vascular
pressures
More recruitment,
distention
Lower resistance
Greater blood flow

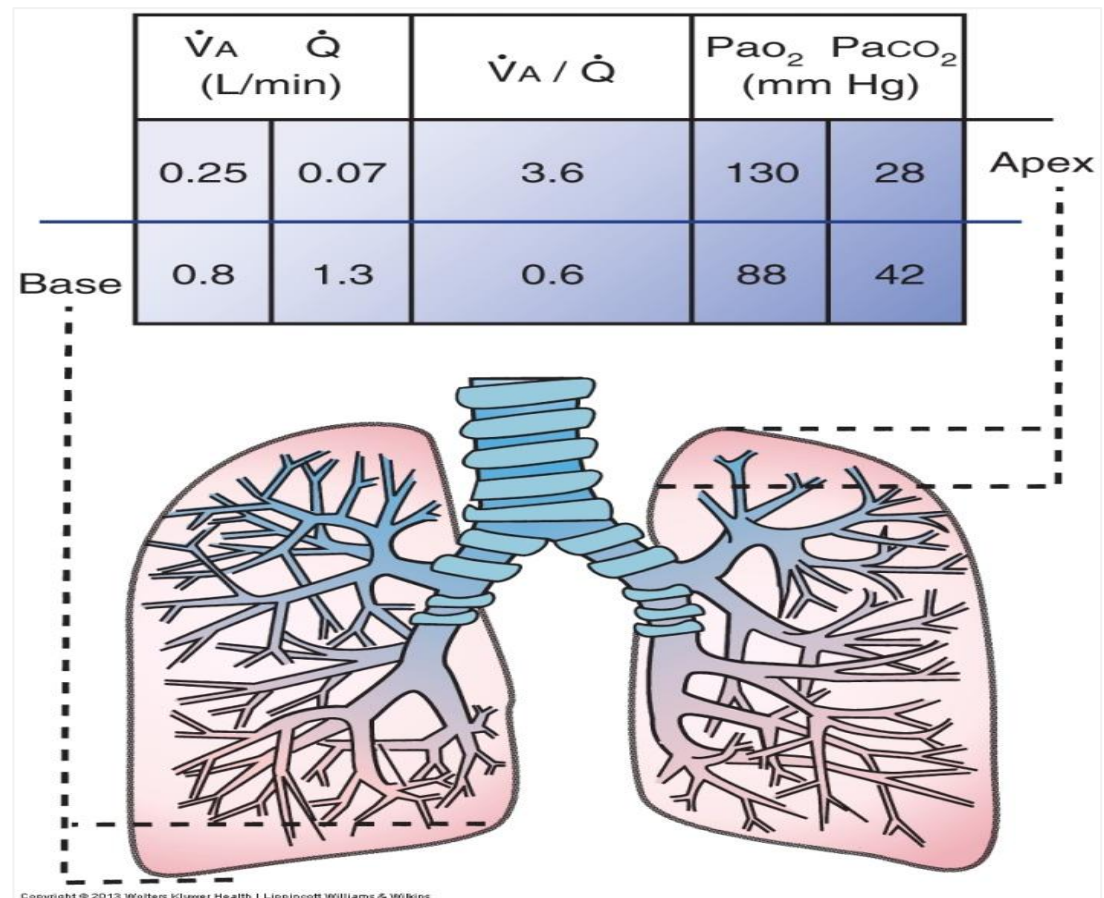
Ventilation Perfusion Ratio Gradient

- Ventilation and blood flow are both gravity dependent.
- Blood flow is proportionally greater than ventilation at the base and vice versa at the apex
- Ventilation-perfusion ratio decreases down the lung



Ventilation/perfusion ratios (V_A/Q) and Capillary Blood gas tension.

Gas exchange is more efficient at the apex of the lung as compared to the base.



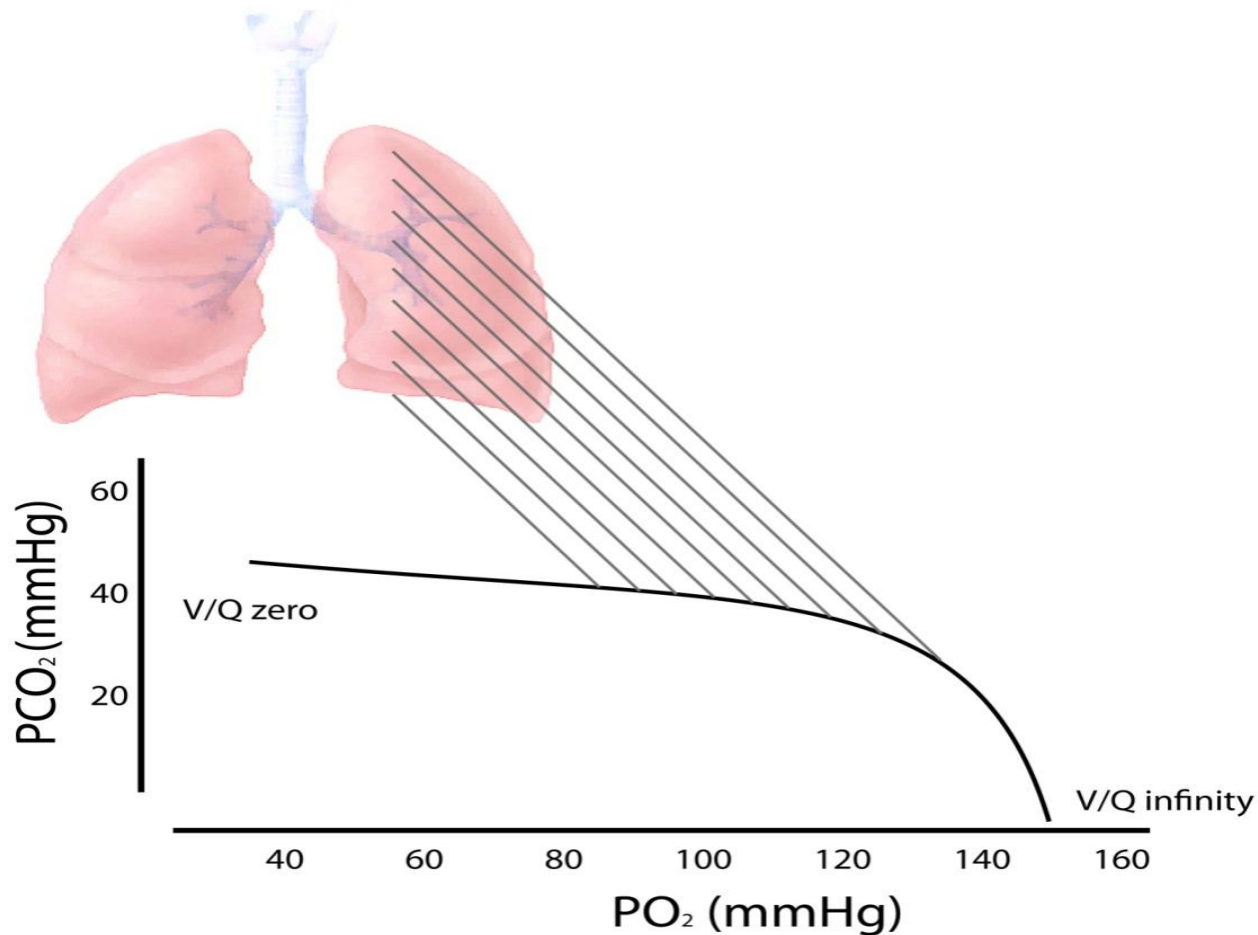
Summary I

- The V_A/Q ratio compares alveolar ventilation with blood flow in lung regions.
- Ventilation and blood flow are both gravity dependent with airflow and blood flow increasing down the lung resulting in variations in V_A/Q
- Ventilation and Perfusion are higher at the base as compared to the apex
- V_A/Q ratio is high at the apex and low at the base
- When blood flow exceeds alveolar ventilation (as seen at the base), a fraction of the blood passes through the pulmonary capillaries at the base of the lungs without becoming fully oxygenated.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0248

Predict how the presence of abnormally low and high V/Q ratios in a person's lungs will affect arterial PO₂ and PCO₂.

V/Q ratio and Regional Blood Gases



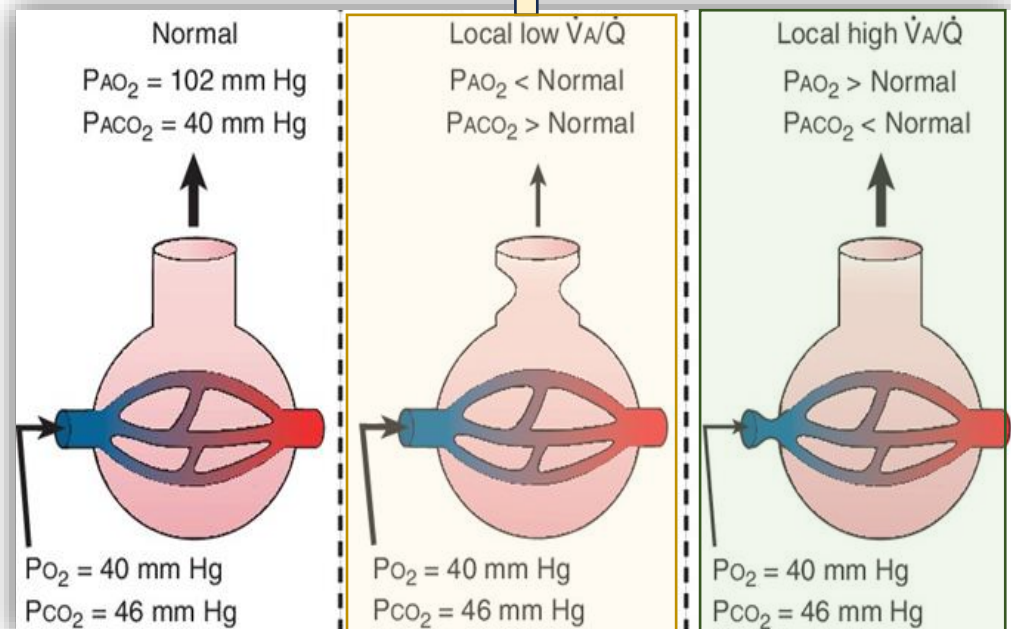
Lung Location and V/Q ratio

Location	Lung Vol (%)	V/Q	PO ₂ (mm Hg)	PCO ₂ (mm Hg)	pH	Q
Apex	7	3.3	130	28	7.55	0.07
Base	13	0.6	88	42	7.38	1.3
Overall	100	0.84	100	40	7.4	5.0

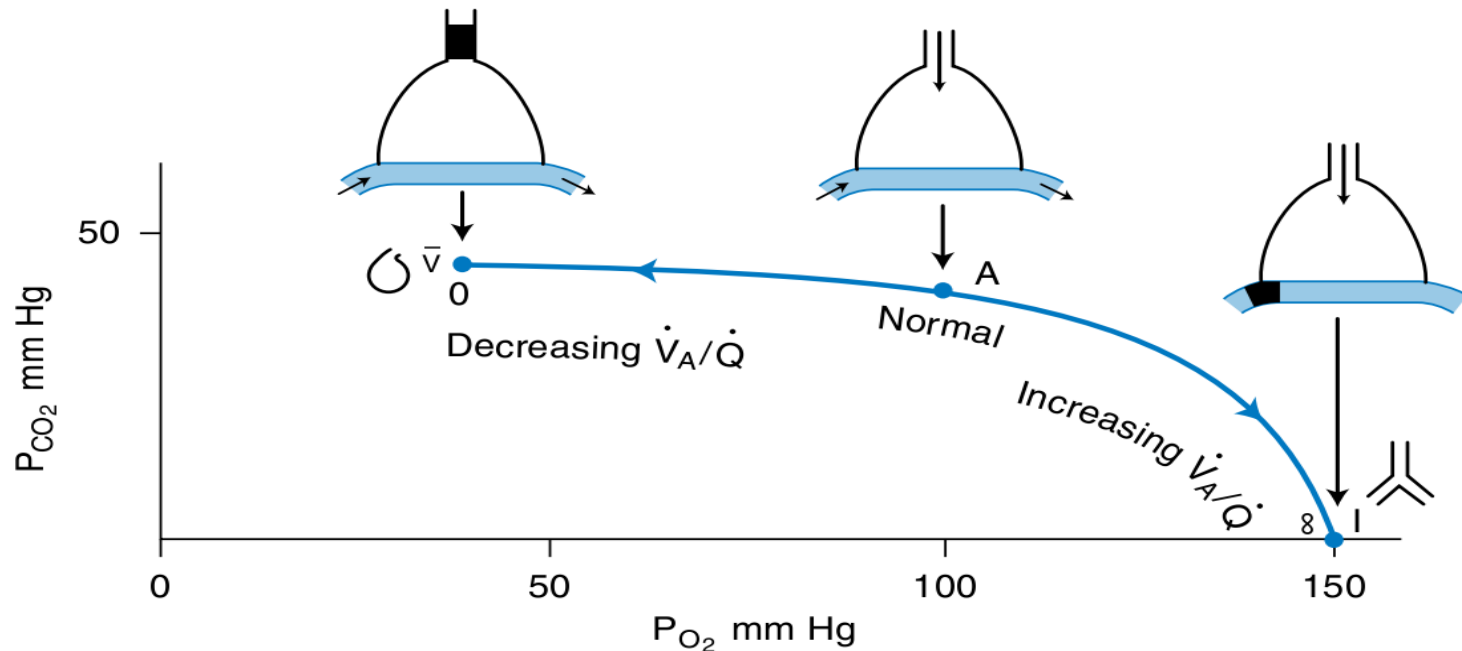
Effect of alteration of V/Q ratio on P_{O_2} and P_{CO_2}

A partially blocked airway causes this region to be underventilated relative to blood flow. A low regional ratio causes venous admixture and will increase the physiologic shunt.

A partially obstructed pulmonary arteriole will cause an abnormally high ratio in a lung region. This region is overventilated leading to an increase in physiologic dead space.



Effect of alteration of V/Q ratio on P_{O_2} and P_{CO_2}



O_2 - CO_2 diagram showing a V/Q ratio line

$\downarrow V/Q = \downarrow PO_2 \uparrow PCO_2$ whilst $\uparrow V/Q = \uparrow PO_2 \downarrow PCO_2$

SOM.MK.II.BPM1.3.CPR.2.PHYS.0251

List and explain the **airway and vascular control mechanisms** that help maintain a normal **ventilation/perfusion ratio**. Name two compensatory reflexes for V/Q inequality.

Normalizing the V/Q ratio

1. **Hypoxic vasoconstriction:** For low V/Q ratio (lots of blood or too little ventilation); causes the blood coming into the area to be directed to other parts of the lung.
2. **Bronchoconstriction:** for high V/Q ratio, the bronchi will constrict slightly to increase the resistance and decrease the amount of ventilation coming into an area that is not well perfused thus limiting the amount of alveolar dead space

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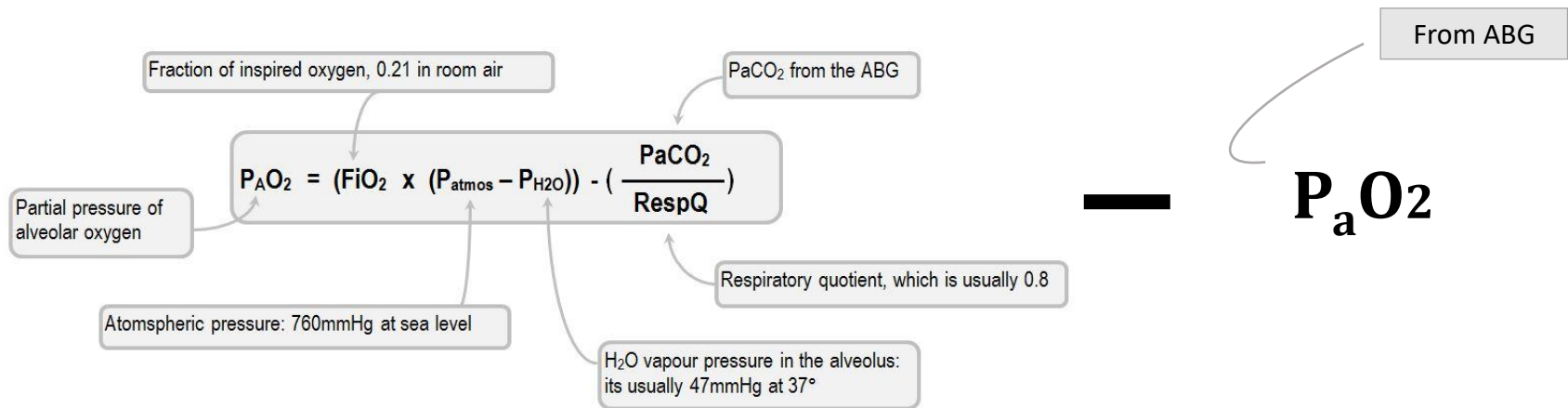
Calculate the alveolar to arterial PO₂ difference, **(A-a) O₂ gradient**. List the **normal value range for (A-a) O₂** and explain the **significance of an elevated (A-a) O₂**.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0253

Describe how **changes to the A-a gradient** occur and what factors lead to an increased A-a gradient in patients.

Measurement of A-a O₂ Gradient

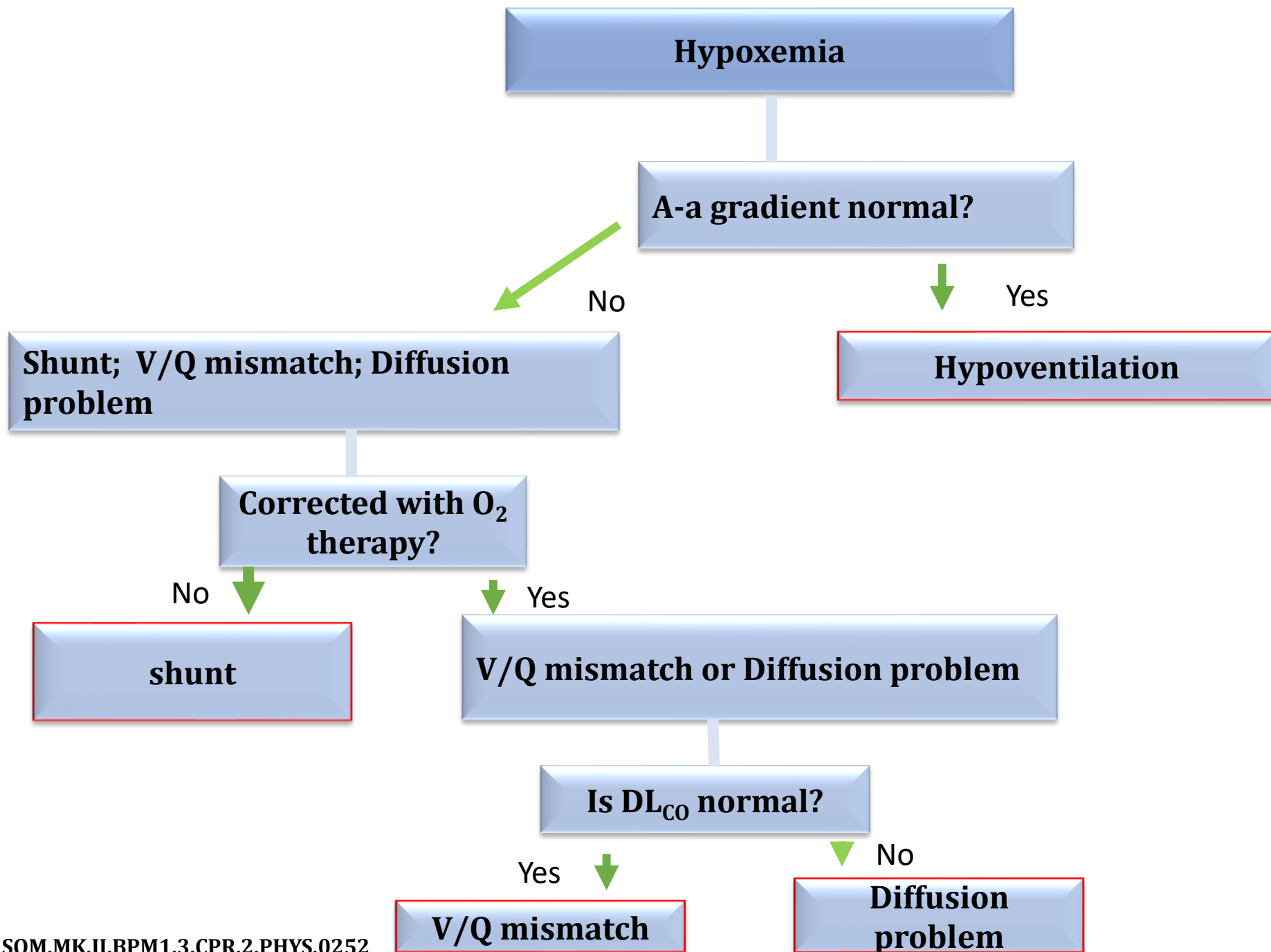
- P_AO₂ without diffusion problems or V_A/Q mismatch should ideally be equal to the P_aO₂
- Normal value 5-15 mm Hg, this increases with age.
- **A-a O₂ Gradient = P_AO₂ - P_aO₂**



Respiratory Causes of Hypoxemia

Causes	Effect on A-a O ₂ Gradient
Regional low V_A/Q ratio (90%)	↑
Anatomic shunt	↑
Generalized Hypoventilation	—
Diffusion block	↑

Respiratory causes of hypoxemia and its effects on A-a O₂ Gradient



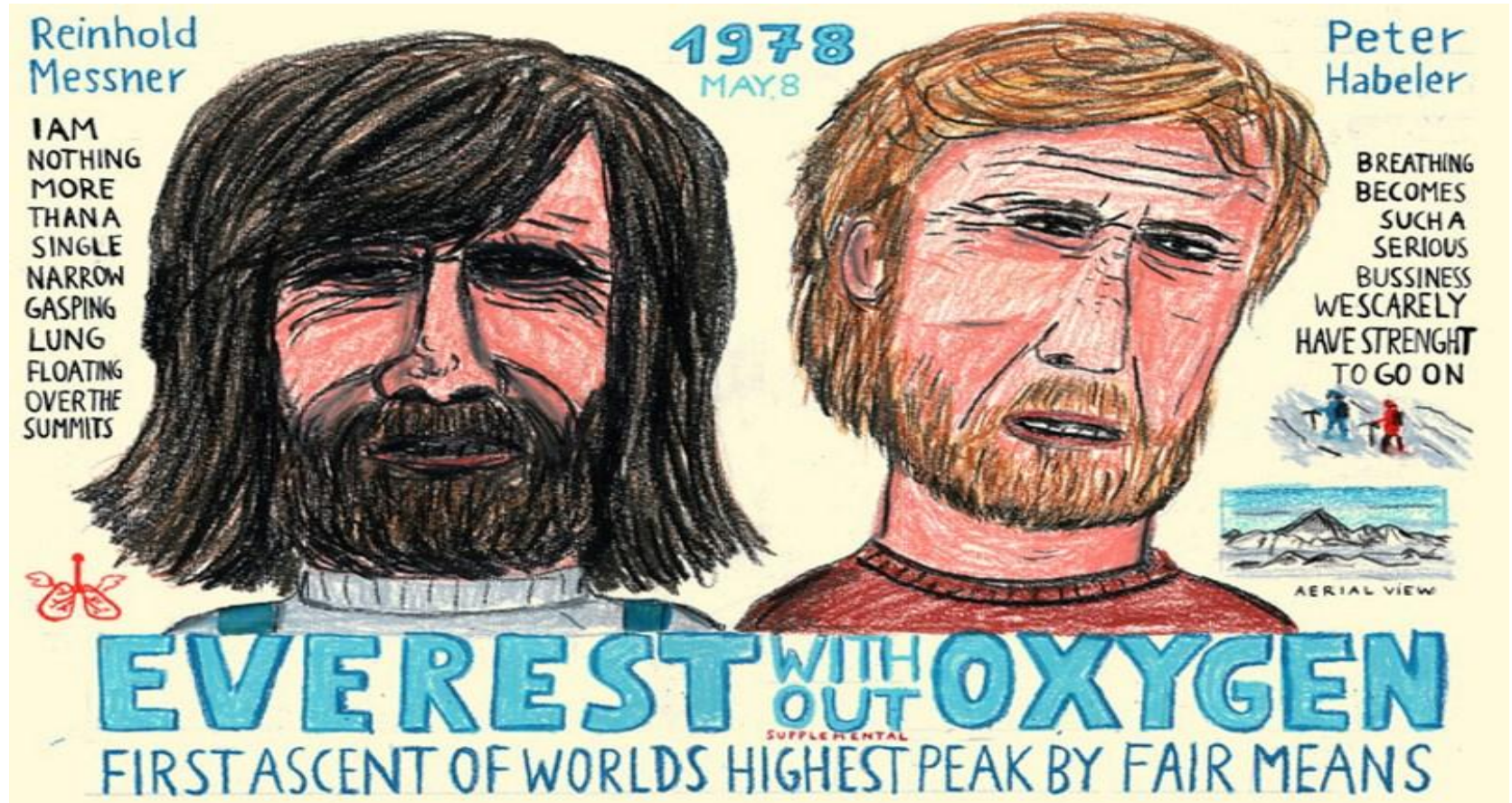
Summary II

- Any deviation of V/Q from the ideal will impact gas exchange
- $V/Q = 0$ = perfusion of an area with no ventilation i.e. a shunt
- $V/Q = \text{infinity}$ = dead space – ventilation with no perfusion

SOM.MK.II.BPM1.3.CPR.2.PHYS.0254

Describe the mechanisms for the shift in **alveolar ventilation** that occur immediately upon **ascent to high altitude**, after remaining at altitude for two weeks, and immediately upon return to sea level.

Altitude and Acclimation



<http://hikingexperience.gr/mount-everest-first-ascent-without-oxygen/>

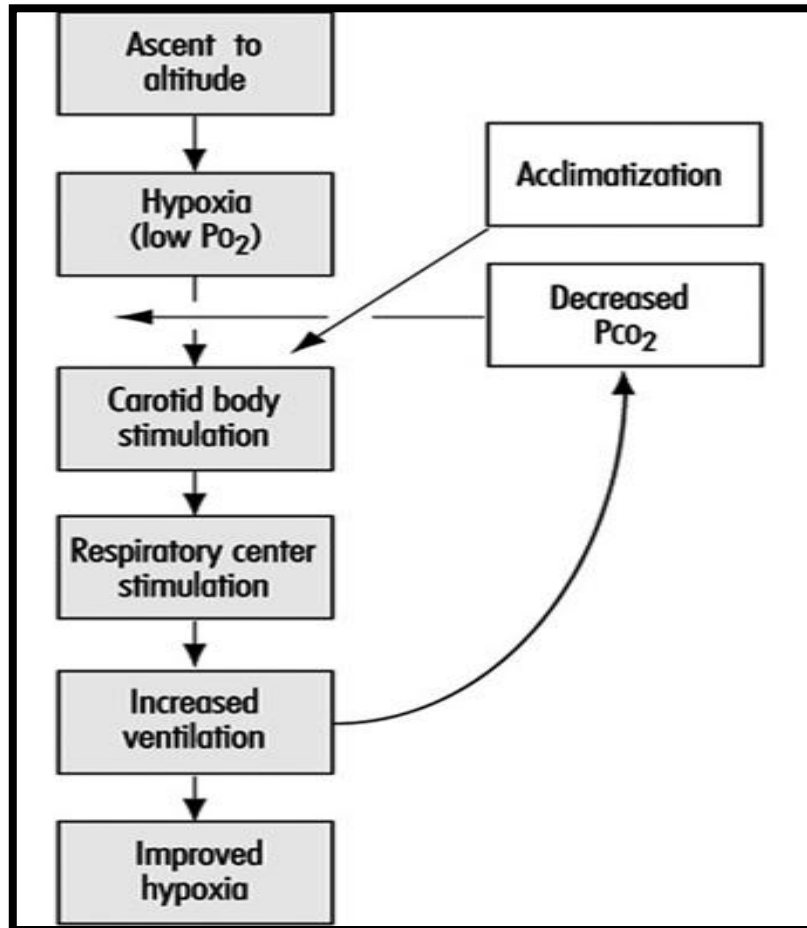
Altitude Physiology

- Three altitude regions that reflect the lowered PO_2 in the atmosphere that can stimulate a ventilatory response :
 1. high altitude (1,500 to 3,500 m or 4,900 to 11,500 ft),
 2. very high altitude (3,500 to 5,500 m or 11,500 to 18,000 ft),
 3. extreme altitude (above 5,500 m, 18,000 ft)
- Altitudes at which PO_2 is insufficient to sustain human life is referred to as the **death zone** (8,000 m or 26,000 ft).
- **Acclimatization**: also called acclimation is the process in which an individual organism adjusts to a change in its environment. Such responses are often reversible should environmental conditions revert to an earlier state.

Ventilatory Response to Altitude – Acute Response

- Low PO_2 causes hypoxia-induced stimulation via the carotid bodies
- This early increase in ventilation seen is small compared with that seen after 8 hours.
- An immediate rise in cardiac output occurs.
- After 8 hours of chronic hypoxia, minute ventilation levels off and is sustained.
- The small rise in ventilation seen in this is due to the significant decrease in $Paco_2$ as a result of excess carbon dioxide blown off with altitude-induced hyperventilation.

Ventilatory Response to Altitude – Acute Response



- The hypoxic stimulation is strongly inhibited by hypocapnia (low PaCO₂).
- The hypocapnia and the rise in blood pH strongly blunt the hypoxic drive.

How does Ventilation increase further?

- The kidneys excrete bicarbonate and the pH is lowered
- Carotid bodies become more sensitive to PO₂

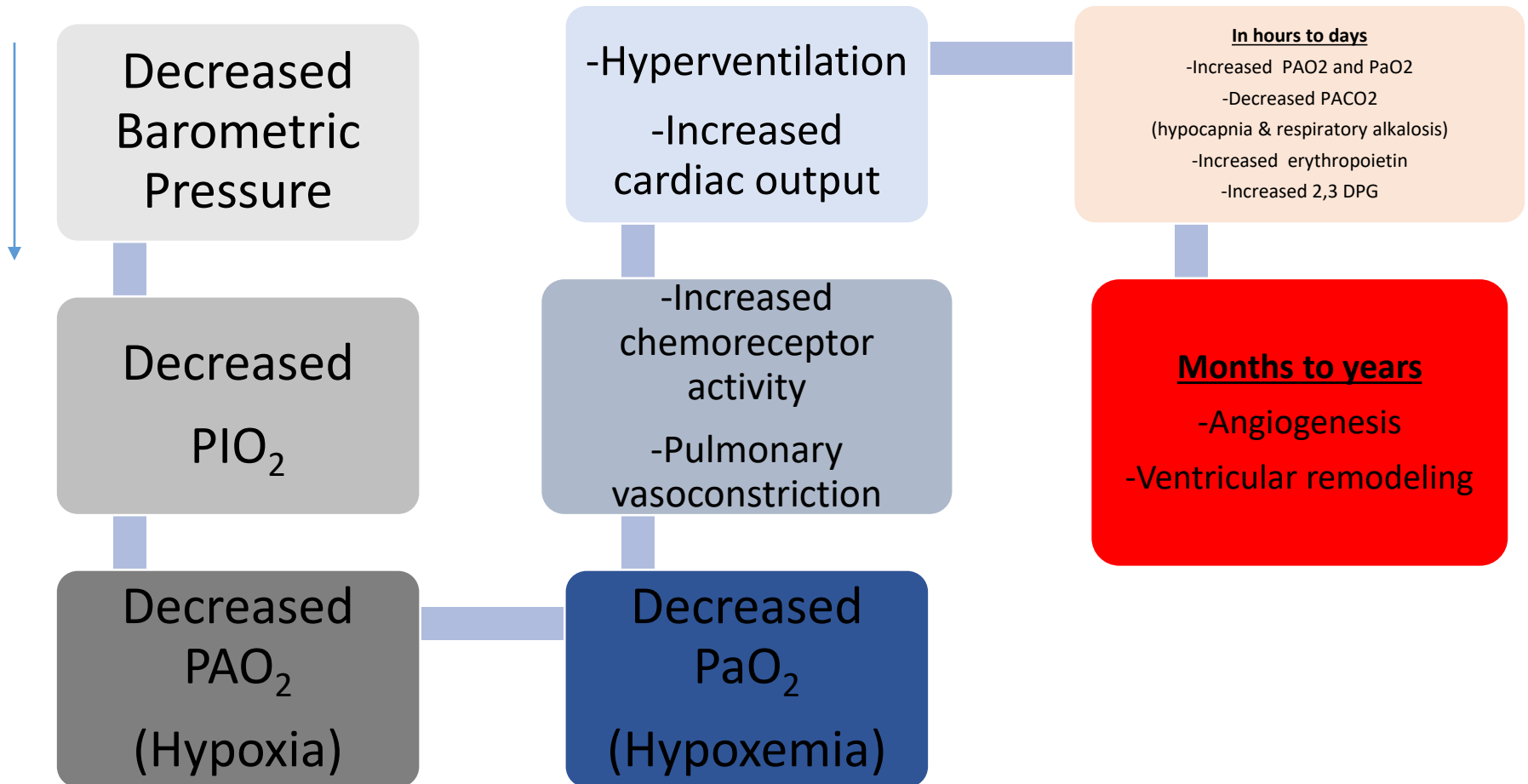
Ventilatory Response to Altitude – Chronic Response

- The second stage of chronic hypoxia-induced hyperventilation is referred to as ventilatory-induced acclimatization.
- This occurs over days of continuous exposure to hypoxia and reaches a stable plateau after 2 weeks.
- This is a functional state that can be reversed if the altitude changes.

Ventilatory Response to Altitude – Chronic Response

- Polycythemia – low PO_2 of kidney stimulates the release of erythropoietin – which in turn stimulates bone marrow to increase RBC production
- Hypoxemia – induces a higher RBC count per unit volume – i.e. more Hb dl^{-1} , i.e. greater carrying capacity for oxygen
- Blood viscosity is greatly increased
- 2-3 DPG levels rise resulting therefore in increased oxygen unloading

Control of Respiration at Low O₂ Tensions: High Altitude



Mountain Sickness: Pulmonary effects

- The pulmonary hypoxic vasoconstriction increases pulmonary vascular resistance – increasing hydrostatic pressure in the pulmonary circulation
- Cardiac output is also increased – increased sympathetic stimulation due to arterial chemoreceptor stimulation
- These factors favor pulmonary edema
- Analysis of pulmonary exudate demonstrates high protein levels – i.e. capillary permeability is increased

Mountain Sickness: Cerebral Effects

- Cerebral circulation is sensitive to hypocapnia – it is a strong cerebral vasoconstrictor
- At altitude the brain receives blood flow with low PO_2 this may be compounded by reduced blood flow due to cerebral hypocapnic vasoconstriction
- Headaches, mental confusion etc. common symptom of altitude sickness
- Ultimately though – low PO_2 leads to cerebral vessel vasodilation and hyperperfusion of the vessels
- Increases the likelihood of cerebral edema

Ventilatory Response to Altitude – Returning to Sea Level

- The ventilatory-induced acclimatization can be reversed with return to sea level.
- Reverse altitude sickness, known as high-altitude de-acclimatization syndrome (HADAS) can occur

Summary

- Acclimatization results in restoration of oxygen delivery towards sea-level values.
- At high altitude ventilation is stimulated by hypoxemia, the resultant hyperventilation counters the full hypoxic ventilatory drive for 2-3 days
- Long term changes include: increase 2-3 DPG, erythropoietin and RBCs